**IT4010 Research Project – 2025 July**

**Topic Assessment Form**

**Project ID :** 25-26J-265

1. Topic (12 words max)

GoodRoad - Citizen-Powered Traffic Violation Reporting System

2. Research group the project belongs to

**CoEAI - Centre of Excellence for AI**

3. Specialization of the project belongs to

**Information Technology (IT)**

4. If a continuation of a previous project:

| | |
|---|---|
| Project ID | |
| Year | |

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Traffic violations, such as reckless driving, red-light running, and illegal parking, contribute significantly to global road accidents and congestion, causing thousands of fatalities and substantial economic losses annually. Law enforcement agencies often lack the resources to monitor and enforce traffic regulations comprehensively, leading to under-enforcement. Traditional enforcement relies on manual patrols or fixed cameras, which are limited by manpower, coverage, and high costs, missing dynamic violations like illegal lane changes [1]. Emerging crowdsourced systems, such as Waze, enable community reporting but lack video evidence or automated detection, while government-operated AI solutions, like automated license plate recognition, are not citizen-driven and omit incentive mechanisms [2].

The increasing adoption of dashcams among drivers presents an opportunity for a citizen-powered traffic violation reporting system. However, no scalable platform exists that leverages citizen-submitted dashcam footage, automates violation detection, and incentivizes participation through rewards. Key gaps include: (1) ensuring the authenticity and quality of submitted videos for legal admissibility, (2) developing AI/ML models to detect violations and identify vehicles in diverse conditions (e.g., night, rain), (3) integrating with government systems for fine collection, and (4) dynamically optimizing incentives for fairness and sustainability [3]. These challenges require advanced AI techniques, including computer vision for video validation and violation detection, deep learning for license plate recognition, and reinforcement learning for reward optimization.

This research investigates the feasibility of a Citizen-Powered Traffic Violation Reporting System with four modular components: video submission and validation, violation detection, vehicle identification, and reward distribution. The study aims to address these gaps by exploring state-of-the-art AI/ML/DL methods, advancing human-centered civic technology, and ensuring compliance with legal frameworks. By enabling scalable, citizen-driven enforcement, the CPTVRS could enhance road safety and reduce congestion, offering a novel contribution to AI-driven civic solutions.

**References:**

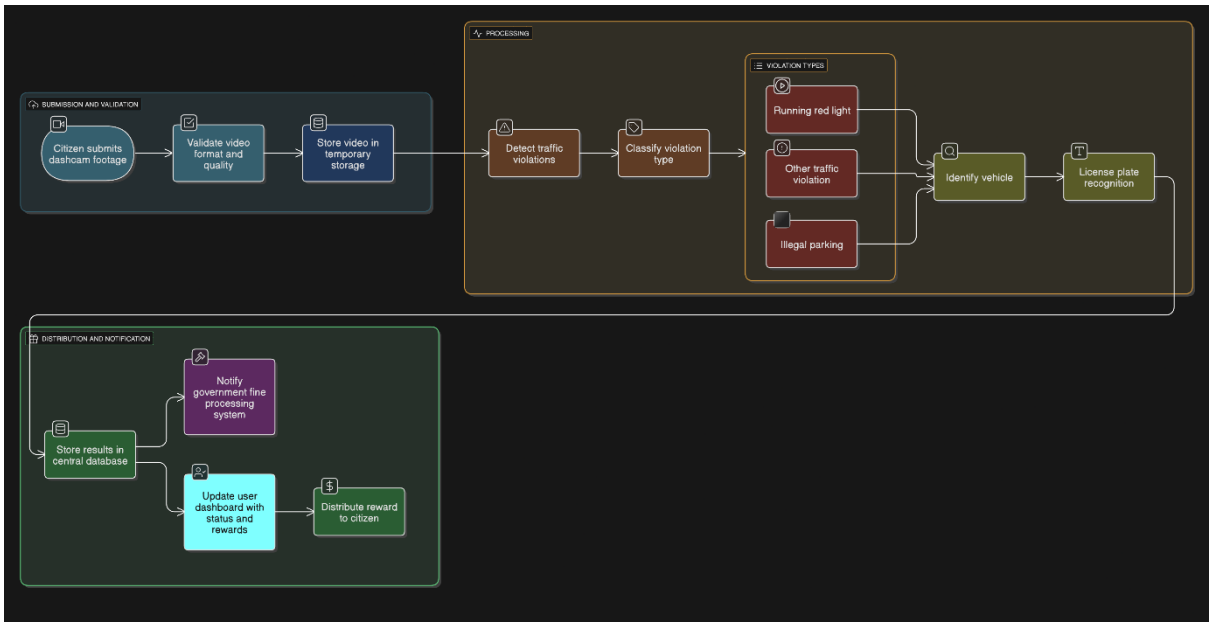
[1] World Health Organization. (2023). *Global status report on road safety 2023*. Geneva: WHO.

[2] Smith, R., & Jones, T. (2021). Crowdsourcing for urban mobility: A review of traffic reporting systems. *Transportation Research Part C*, 130, 103289.

[3] Lee, J., Kim, H., & Park, S. (2022). AI-based traffic enforcement: Challenges and opportunities. *Journal of Intelligent Transportation Systems*, 26(3), 123–135.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The GoodRoad - Citizen-Powered Traffic Violation Reporting System aims to research a scalable platform that leverages citizen-submitted dashcam footage to enhance traffic violation enforcement using AI/ML/DL techniques. The system comprises four modular components: (1) **Video Submission and Validation**, where citizens upload footage via a web/mobile interface, and AI (e.g., CNN) validates quality and authenticity; (2) **Violation Detection**, employing computer vision (e.g., object detection models) to identify and classify violations like red-light running; (3) **Vehicle Identification**, using deep learning (e.g., CRNN) to recognize license plates and link to vehicle registries; and (4) **Reward Distribution**, applying reinforcement learning to optimize commission rates for submitters, integrating with payment systems. Each component operates independently, ensuring modularity and research flexibility, while collectively addressing gaps in automated, citizen-driven enforcement. The platform will explore integration with government systems for fine collection and legal compliance, promoting transparency via user dashboards. This research focuses on feasibility, model development, and system design, advancing civic technology for road safety.




7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The GoodRoad - Citizen-Powered Traffic Violation Reporting System (CPTVRS) requires specialized expertise in **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Deep Learning (DL)**, focusing on computer vision and reinforcement learning. Experts in **Convolutional Neural Networks (CNNs)** are needed for video validation to detect tampering and ensure quality, while proficiency in **object detection models** (e.g., YOLO) is essential for identifying traffic violations like red-light running. Knowledge of **Character Recognition Neural Networks (CRNNs)** is critical for license plate recognition under diverse conditions. **Reinforcement Learning (RL)** expertise, such as Q-learning, is required to optimize dynamic reward systems for citizen incentives. Additionally, **Human-Centered Civic Technology** expertise ensures user-friendly interfaces and legal compliance, necessitating collaboration with legal professionals to align with evidence admissibility standards.

**Software engineering** skills in web/mobile development (e.g., React, Django) and database management (e.g., PostgreSQL) are vital for platform scalability. Knowledge of API integration (e.g., payment gateways) supports fine collection and reward distribution. Familiarity with **data privacy** and cybersecurity ensures secure handling of citizen-submitted footage.

Data requirements include: (1) **dashcam footage datasets** for training CNNs and object detection models, including tampered/untampered videos (e.g., DeepFake Detection Challenge); (2) **traffic violation datasets** (e.g., KITTI) for violation classification; (3) **license plate datasets** (e.g., CCPD) for CRNN training; and (4) **historical fine and participation data** for RL reward optimization. Synthetic data may be generated to address dataset gaps. Access to government vehicle registries (mock or real) and legal frameworks is necessary for system integration and compliance.

8. Objectives and Novelty

Main Objective

The GoodRoad - Citizen-Powered Traffic Violation Reporting System aims to investigate the development of a scalable web and mobile application that enables citizens to submit dashcam footage of traffic violations for automated processing, verification, and reward distribution. The system comprises four independent components, each incorporating AI/ML/DL for research significance, ensuring modularity and advancing civic technology.

| Member Name with Registration No | Sub Objective | Tasks | Novelty |
|---|---|---|---|
| Wijerathna P. G. S. P.<br>IT22148872 | Video Submission and Validation Module | • Develop frontend interface (web and mobile form) for video submission<br>• Build backend API for handling submissions<br>• Implement validation system for video format, duration, and metadata<br>• Create notification system for submission confirmation | • Researching CNN-based models for video quality validation and tampering detection using datasets like DeepFake Detection Challenge<br>• Addresses gap in automated video validation for dashcam footage, enhancing evidence reliability and scalability |
| Liyanage R. L. N. L.<br>IT22151438 | Violation Detection Module | • Implement video retrieval from database<br>• Develop violation detection techniques (e.g., YOLO for vehicle detection and classification) | • Exploring YOLO-based object detection models trained on traffic violation datasets for real-time processing<br>• Advances enforcement beyond static camera systems |

| | | | |
|---|---|---|---|
| | | • Generate reports with timestamps and confidence scores<br>• Design storage strategy for results | with scalable, automated violation detection |
| Rathnayake W. K. D.<br>IT22174444 | Vehicle Identification Module | • Process video frames for analysis<br>• Develop license plate detection methods<br>• Build OCR pipeline for plate number recognition<br>• Integrate with vehicle registries via database | • Investigating CRNN models for license plate recognition using datasets like CCPD and synthetic plates<br>• Enhances automated vehicle identification with customized OCR for diverse license plate formats |
| Hendawitharana H. W. K. P.<br>IT22219466 | Reward Distribution and Reporting Module | • Calculate commissions based on violation severity and fine recovery<br>• Integrate payment systems for fund transfers<br>• Develop dashboard for submission status and earnings | • Exploring reinforcement learning models (e.g., Q-learning, DQN) for dynamic commission optimization using historical data<br>• Novel application in civic tech to maximize engagement and sustainability through dynamic rewards |

9. Individual component description of how it is complied with the specialization.

| Member Name with Registration No | Description |
|---|---|
| | |
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10. Supervisor details

| | Title | First Name | Last Name | Signature |
|---|---|---|---|---|
| Supervisor | Dr. | Mahima | Weerasinghe | |
| Co-Supervisor | Dr. | Lakmini | Abeywardhana | |
| External Supervisor | | | | |
| Summary of external supervisor's (if any) experience and expertise | | | | |

**This part is to be filled by the Topic Screening Staff members.**

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

| Yes | | No | |
|---|---|---|---|

b) Does the proposed topic exhibit novelty?

| Yes | | No | |
|---|---|---|---|

c) Do you believe they have the capability to successfully execute the proposed project?

| Yes | | No | |
|---|---|---|---|

d) Do the proposed sub-objectives reflect the students' areas of specialization?

| Yes | | No | |
|---|---|---|---|

e) Supervisor's Evaluation and Recommendation for the Research topic:

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| |

| IT4010 Research Project – 2025 July<br><br>Topic Assessment Form |
|---|

Acceptable: Mark/Select as necessary

| | |
|---|---|
| Topic Assessment Accepted | |
| Topic Assessment Accepted with minor changes* | |
| Topic Assessment to be Resubmitted with major changes* | |
| Topic Assessment Rejected. Topic must be changed | |

* Detailed comments given below

Comments

| |
|---|
| |

| Staff Member's Name | Signature |
|---|---|
| | |
| | |

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.